# Midterm Exam - ECO 3121
# (Introductory Econometrics, Fall 2019)

INSTRUCTION: Write your answers as clearly and concisely as possible, and feel free to use mathematical expressions in your explanations. You can not use a calculator or electronic dictionary, or any books, notes, or scratch paper - all of these items must be placed at the front of the classroom. Having any of these items with you during the exam constitutes cheating. **ALL of your answers (including answers to multiple choice questions) must appear on the answer book, and answers on the exam will not be graded.** The exam consists of **10** multiple choice questions and **4** short answer questions. Please check that you have **8** pages.

PRINTED NAME: _____

STUDENT I.D.: _____

SEAT NO.: _____

Good luck and enjoy the exam!

**I. (30 points) Multiple Choice. Choose the one alternative that best answers the question.**

1. Studying unemployment in Portugal from 1966 to 2009 is an example of using
   (a) Cross-sectional data.
   (b) Time series data.
   (c) Panel data.
   (d) Experimental data.

2. Consider the following two statements: I) If two random variables $X$ and $Y$ are uncorrelated, then they are independent. II) If two random variables $X$ and $Y$ are independent, then they are uncorrelated.
   (a) Only II) is true.
   (b) Only I) is true.
   (c) Both I) and II) are true.
   (d) Both I) and II) are false.

3. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean $\mu_Y$. Unbiasedness of the sample average $\bar{Y}$ means
   (a) the probability that $\bar{Y}$ is close to $\mu_Y$ increases as the sample size increases.
   (b) the mean of the sampling distribution of $\bar{Y}$ is zero.
   (c) $\bar{Y}$ will be equal to $\mu_Y$ if the sample size is larger.
   (d) if numerous repeated samples of the same size are taken, on average the value of $\bar{Y}$ will be equal to $\mu_Y$.

4. A $p$-value smaller than the significance level implies
   (a) a small t-statistic.
   (b) a small standard error.
   (c) the rejection of the null hypothesis.
   (d) that the Central Limit Theorem is not appropriate.

5. If the three OLS assumptions hold, which is not the property of the OLS estimator $\hat{\beta}_1$?
   (a) $\hat{\beta}_1$ is unbiased.
   (b) $\hat{\beta}_1$ is consistent.
   (c) $\hat{\beta}_1$ is equal to $\beta_1$ with probability one.

(d) $\hat{\beta}_1$ is asymptotically normal.

6. The error term is homoskedastic if

   (a) $var(u|X = x)$ is constant.

   (b) there are not outliers.

   (c) $u$ is normally distributed.

   (d) $var(u|X = x)$ depends on $x$.

7. Suppose we include $X_1$ in our single regression, but we leave out $X_2$, a variable that is negatively correlated with the included variable $X_1$. $X_2$ is a determinant of $Y$ (the dependent variable), and the effect of $X_2$ on $Y$ is positive. The effect of $X_1$ on $Y$ is negative. What is the consequence of leaving $X_2$ out of the model?

   (a) The OLS estimator of the effect of $X_1$ on $Y$ will be consistent.

   (b) The OLS estimator of the effect of $X_1$ on $Y$ will be positive.

   (c) The OLS estimator of the effect of $X_1$ on $Y$ will be zero.

   (d) The OLS estimator of the effect of $X_1$ on $Y$ will be too negative (negative bias).

8. What is the relation between the standard error of $\hat{\beta}_1$ (denoted as $SE(\hat{\beta}_1)$) and the variance of $\hat{\beta}_1$ (denoted as $Var(\hat{\beta}_1)$)?

   (a) $SE(\hat{\beta}_1)$ is equal to the square root of $Var(\hat{\beta}_1)$.

   (b) $SE(\hat{\beta}_1)$ is equal to the square root of an estimator of $Var(\hat{\beta}_1)$.

   (c) $SE(\hat{\beta}_1)$ is equal to an estimator of $Var(\hat{\beta}_1)$.

   (d) $SE(\hat{\beta}_1)$ is equal to $Var(\hat{\beta}_1)$.

9. To consider whether gender matters in terms of earnings, a student defines two dummy variables $M_i = 1$ if the person is a male and 0 otherwise, and $F_i = 1$ if the person is a female and 0 otherwise. Let $Y_i$ be the person's earnings. The student can run all the following regression models (and obtain estimates for the coefficients) except

   (a) $Y_i = \beta_0 + \beta_1 M_i + u_i$.

   (b) $Y_i = \beta_0 + \beta_1 F_i + u_i$.

   (c) $Y_i = \beta_0 + \beta_1 M_i + \beta_2 F_i + u_i$.

   (d) $Y_i = \beta_1 M_i + \beta_2 F_i + u_i$.

10. Including an additional regressor in a regression

    (a) always leads to more accurate estimates of regression coefficients.

(b) always reduces the sum of squared residuals.

(c) is advisable only if the variable's t-statistic is larger than 1.96 in absolute value.

(d) increases imperfect multicollinearity and is not advisable

**II. (70 points) Short Answer. Make sure to show your work and explain your steps for full credit.**

1. (10 points) Answer the following questions.

   (a) Suppose $\{Y_i\}_{i=1}^n$ is a i.i.d. sequence of random variables with mean $\mu_Y$ and variance $\sigma_Y^2$. Calculate $Var(\bar{Y})$ and its limit as $n \to \infty$.

   (b) List the four assumptions for multiple linear regression (as described in class or the book).

2. (35 points) You have data on 480 students from CUHK(SZ) and plan to investigate how the GPA varies with the studying hours per week and the majors. In the data there are four types of majors: economics, finance, accounting, and marketing. Each one of the 480 students exclusively belongs to one of the four majors (there is no double major). In the data there are four dummy variables: $ECON = 1$ if economics major and 0 otherwise; $FIN = 1$ if finance major and 0 otherwise; $ACCT = 1$ if accounting major and 0 otherwise; $MAK = 1$ if marketing major and 0 otherwise;. The regression results are

$$\widehat{GPA} = \underset{(1.23)}{1.81} + \underset{(0.02)}{0.06} \times SH + \underset{(0.04)}{0.15} \times ECON + \underset{(0.03)}{0.10} \times FIN + \underset{(0.02)}{0.08} \times MAK$$
$$R^2 = 0.69, RMSE = 4.65$$

   where $SH$ is the studying hours per week and the values in parentheses are robust standard errors. In the data, $SH \geq 5$.

   (a) Interpret the estimated coefficients on $SH$ and $ECON$.

   (b) How do you interpret the intercept? Does it make sense?

   (c) Is the effect of $SH$ significantly different from zero at the 5% level?

   (d) Why did you exclude the dummy variable $ACCT$ in the regression?

   (e) If you excluded all dummy variables, i.e., regress $GPA$ on just constant and $SH$, what would be the change of $R^2$ and $RMSE$? Smaller or larger?

   (f) Represent in a graph the implicit GPA function for the four majors. (The estimated GPA in the $y$ axis and the studying hours per week $SH$ in the $x$ axis).

(g) Suppose you wish to test whether finance major students' GPA is significantly different from other majors and rerun the regression by including $ACCT$ and excluding $FIN$. Without using the stata, you can still get the estimated coefficients and fill the blanks and explain your answer.

$$\widehat{GPA} = \square + \square \times SH + \square \times ECON + \square \times ACCT + \square \times MAK,$$
$$R^2 = \square,\ RMSE = \square.$$

3. (10 points) Suppose that an individual's salary (*salary*) depends on his/her education (*edu*) and unobserved factors that affect his/her salary: $salary = \beta_0 + \beta_1 edu + u$. Do you expect this model to satisfy the first OLS assumption $E(u|edu) = 0$? Explain your answer. If not, what happen to the OLS estimator of $\beta_1$ (the direction of the bias)?

4. (15 points) Answer the following two questions.

   (a) Consider the simple linear regression model without intercept: $Y_i = \beta_1 X_i + u_i, i = 1, \ldots, n$ . It is known that the OLS estimator of $\beta_1$ is given by $\hat{\beta}_1 = \sum_{i=1}^n X_i Y_i / \sum_{i=1}^n X_i^2$. Let $\hat{Y}_i = X_i \hat{\beta}_1$, and define $R_u^2 = \sum_{i=1}^n \hat{Y}_i^2 / \sum_{i=1}^n Y_i^2$. Prove that $0 \leq R_u^2 \leq 1$.

   (b) Consider the multiple linear regression model: $Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i + u_i, i = 1, \ldots, n$ . Let $\hat{\beta}_1$ and $\hat{\beta}_2$ be the OLS estimator of $\beta_1$ and $\beta_2$, and their standard errors be given by $SE(\hat{\beta}_1)$ and $SE(\hat{\beta}_2)$. Let $\hat{\Delta}$ be a consistent estimator of the asymptotic covariance of $\sqrt{n}\hat{\beta}_1$ and $\sqrt{n}\hat{\beta}_2$ (that is, the limit of the covariance of $\sqrt{n}\hat{\beta}_1$ and $\sqrt{n}\hat{\beta}_2$ as $n \to \infty$.) How do you construct a $t$ test for $H_0 : \beta_1 + \beta_2 = 0$ v.s. $H_0 : \beta_1 + \beta_2 > 0$? Please specify the $t$ statistic (in particular, the standard error), and the distribution of the $t$ statistic under $H_0$.

# Appendix A Some Formulas

Probability and Statistics:

- Mean: $\mu_Y = E[Y]$; Variance: $\sigma_Y^2 = E[(Y-\mu_Y)^2]$;
  Skewness: $E[(Y-\mu_Y)^3]/\sigma_Y^3$; Kurtosis: $E[(Y-\mu_Y)^4]/\sigma_Y^4$
- Law of iterated expectation: $E[Y] = E[E[Y|X]]$
- $\bar{Y} = n^{-1}\sum_{i=1}^n Y_i$, $s_Y^2 = (n-1)^{-1}\sum_{i=1}^n (Y_i-\bar{Y})^2$, $SE(\bar{Y}) = s_Y/\sqrt{n}$
- $cov(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$, $s_{X,Y} = (n-1)^{-1}\sum_{i=1}^n (X_i-\bar{X})(Y_i-\bar{Y})$
- LLN: If $Y_1,\ldots,Y_n$ are i.i.d. and $\sigma_Y^2 < \infty$, then

  $$\bar{Y} \xrightarrow{p} \mu_Y \ ,$$

  which means that for any $\epsilon > 0$, $P(|\bar{Y}-\mu_Y| < \epsilon) \to 1$ as $n \to \infty$.
- CLT: If $Y_1,\ldots,Y_n$ are i.i.d. and $0 < \sigma_Y^2 < \infty$, then

  $$\frac{\sqrt{n}(\bar{Y}-\mu_Y)}{\sigma_Y} \xrightarrow{\text{d}} N(0,1) \ ,$$

  which means that the cdf of $\sqrt{n}(\bar{Y}-\mu_Y)/\sigma_Y$ is convergent to the cdf of $N(0,1)$.

Simple OLS Regression:

- $\hat{\beta}_1 = s_{X,Y}/s_X^2$, $\hat{\beta}_0 = \bar{Y} - \bar{X}\hat{\beta}_1$, $\hat{u}_i = Y_i - \hat{\beta}_0 - X_i\hat{\beta}_1$
- $\sqrt{n}(\hat{\beta}_1 - \beta_1) \xrightarrow{\text{d}} N(0, \sigma_X^{-4} var[(X-\mu_X)u])$, $SE(\hat{\beta}_1) = \sqrt{n^{-1}s_{X\hat{u}}^2/s_X^4}$,
  where $s_{X\hat{u}}^2 = n^{-1}\sum_{i=1}^n (X_i-\bar{X})^2\hat{u}_i^2$
- $SER = \sqrt{(n-2)^{-1}\sum_{i=1}^n \hat{u}_i^2}$, $R^2 = 1 - [(n-2)SER^2]/[(n-1)s_Y^2]$

# Appendix B Distributional Tables

Table 1: Large Sample Critical Values for the $t$-statistic

| | Significance Level | | |
|---|---|---|---|
| | 10% | 5% | 1% |
| 2-Sided Test | 1.64 | 1.96 | 2.58 |
| 1-Sided Test(>) | 1.28 | 1.64 | 2.33 |
| 1-Sided Test(<) | -1.28 | -1.64 | -2.33 |

Table 2: Large Sample Critical Values for the $F$-statistic

| | Significance Level | | |
|---|---|---|---|
| Degrees of Freedom (q) | 10% | 5% | 1% |
| 1 | 2.71 | 3.84 | 6.63 |
| 2 | 2.30 | 3.00 | 4.61 |
| 3 | 2.08 | 2.60 | 3.78 |
| 4 | 1.94 | 2.37 | 3.32 |
| 5 | 1.85 | 2.21 | 3.02 |
| 6 | 1.77 | 2.10 | 2.80 |
| 7 | 1.72 | 2.01 | 2.64 |
| 8 | 1.67 | 1.94 | 2.51 |

This is the end of the exam!